Dr. Scott Vinay

Data Scientist | Machine Learning Engineer

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A senior data scientist with significant experience designing and developing novel machine-learning solutions, particularly regarding time-series, language, and graph data. I am a motivated engineer with ambitions to lead technical teams to solve some of the biggest challenges in society.

Employment

Sibylla AI
London, UK (Remote)

Quantitative Researcher (Machine Learning) — *March 2026 – Apr 2026*

- Contributed to research on ML-driven alpha-research strategies for neutral and directional funds, with a heavy emphasis on deployment-scale software engineering.

BT Group (EE)
Birmingham, UK

Senior Data Scientist (Reinforcement Learning) — *July 2024 – Feb 2026*

- Led the ML decisioning team, and in charge of the development of BT's next-best-action recommender system.
- Developed a tool for using agentic LLMs to automatically generate arbitrary code execution graphs, and compile these in a risk-free way. This enabled the automated creation and analysis of reports
- Improved ad allocation via Bayesian bandit models, increasing early-life ad visibility while improving expected CTR (BT *Brilliant* award).
- Built a collaborative filtering model that was the first contender model in a 2 year project to beat the baseline model, in offline testing improving click-through rate by 10 – 30%. Built training and scoring pipelines in GCP Vertex, Kubeflow, and Dataflow to put this into production.
- Developed novel statistical estimators for offline estimation of the performace of pricing models (academic paper currently under production).

Cash App
Melbourne, Australia

Senior Machine Learning Engineer (Modelling) — *May 2023 – Feb 2024*

- Worked on developing models and systems for the detection of fraud, primarily using unsupervised learning to detect anomalies and identify new types of fraud attacks.
- Took the lead role in adapting our machine learning strategy for the launch of Cash App in a new market, involving co-ordinating different teams and designing and implementing systems to track and modify model performances across markets.
- Identified millions of dollars of fraudulent money movements per month by developing a novel framework for detecting and visualising anomalies in arbitrary sequences of data vectors, which was then used by multiple teams. When implemented for payment data for the purpose of scam detection, this reached a 92% top-1000 precision in production, mainly identifying particularly pernicious and costly scam types.
- Designed and developed a full end-to-end pipeline to enable the testing of models on data without ground truth values. This included automatically sending model results to financial experts for manual review, collecting and parsing those results, publishing Tableau dashboards of results, identifying coherent groups of anomalies, and sending summaries of these to Slack. This enabled the detection of a previously unknown scam type.
- Developed successful models for the identification of money-laundering networks, involving both heuristic approaches based on graph-theoretical concepts, and graphical convolutional autoencoder models, which were successfully used for tracking illegal money movements from source to sink.

Transformative AI
United Kingdom

Senior Machine Learning Researcher — *Oct 2021 – May 2023*

- Worked on the development of deep classification models for the prediction of arrhythmias from electrocardiogram data. Developed intricate multi-stage training pipelines and novel deep-learning algorithms for characterising the shapes of time-series data.
- Lead author on report to FDA detailing our product, as well as lead author on publication submitted to Nature Medicine.

Machine Learning Researcher — *July 2020 – Oct 2021*

- Introduced novel Bayesian-based algorithms to adapt models to an individual patient. This improved test-set F_1 score from 40.1% to 77.7%, and area-under-ROC from 64.8% to 92.7%.
- Produced high-quality reports and papers, which contributed significantly to the winning of research grants worth over a million USD.
- Supervised a teams of external contributors and interns, including producing resources and guides for moving to a cloud-based environment.
- Adapted cutting-edge results from research in bias-reduction in AI to improve generalisability to new databases.

Education

PhD in Quantum Cryptography
Sheffield, United Kingdom

THE UNIVERSITY OF SHEFFIELD *Sept. 2015 - March 2019*

- Thesis on *The Statistics and Security of Quantum Key Distribution.*
- Research included development of both practical proposals for technological protocols and novel mathematical frameworks for efficient statistical analysis of arbitrary networks.
- Supervised by Pieter Kok and Stephano Pirandola.

MPhys in Physics
Coventry, United Kingdom

THE UNIVERSITY OF WARWICK *Oct. 2011 - Jun. 2015*

- 1st class honours with an 84% average grade.
- Dissertation on *Energetics of Knotted Defects in Nematic Liquid Crystals.*
- Supervised by Gareth Alexander.

Publications

Statistical analysis of quantum entangled network generation
Physical Review A

SCOTT E. VINAY AND PIETER KOK *April 2019*

- We developed novel numerical and analytic techniques for the analysis of probabilistic network generation, which we apply to the specific case of quantum communication to prove new security bounds.
- Improved secure communication rates by three orders of magnitude over more simplistic methods.
- Techniques used: Markov chains, Cauchy's residue theorem, Fourier transforms, complex analysis.

Extended analysis of the Trojan-horse attack in quantum key distribution
Physical Review A

SCOTT E. VINAY AND PIETER KOK *April 2018*

- We analysed the effect of a side-channel attack on quantum communication, proving an increased bound on security while relaxing the assumptions previously used.
- Techniques used: Covariance matrices, Krauss-decomposition of channels.

Practical repeaters for ultralong-distance quantum communication
Physical Review A

SCOTT E. VINAY AND PIETER KOK *May 2017*

- We presented a complete protocol for a quantum repeater, and an analysis of its efficiency that takes probabilistic effects into careful consideration. We show that it produces high secret key rates relative to comparable protocols.
- Techniques used: Order statistics, entanglement distillation.

Skills and Personal

Software
- Highly proficient with Pandas, Numpy, Matplotlib, Git, SQL, Snowflake, and machine learning packages including Sklearn, Keras, and Pytorch.
- Comfortable and confident in cloud systems including GCP and AWS, in particular using Vertex and SageMaker.
- Knowledgeable about reinforcement learning for solving complex problems.
- Experience with C, MATLAB, and Java.

Presentational
- Confident at public speaking on complex topics.
- Delivered talks at multiple department symposia, as well as at meetings of the multi-university White Rose group.
- Presented research at conferences at the University of Cambridge (UK), Technische Universität Wien (Vienna), and the University of Nottingham (UK).

Personal
- In my personal time I like to keep active through golf and mountain biking.
- I am a keen chess player, and enjoy running a weekly D&D game as game master.